INTER-VEHICULAR NETWORKING

PROJECT PRELIMINARY REPORT

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MAR BASELIOS COLLEGE OF ENGINEERING AND TECHNOLOGY
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Submitted in partial fulfilment of the requirements of the award of Bachelor of Technology Degree in Electrical and Electronics Engineering of the APJ Abdul Kalam Technological University (KTU)

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CERTIFICATE

This is to certify that this report entitled "**INTER-VEHICULAR NETWORKING**" *is a bonafide record of the project preliminary presented by* **ANANTHAKRISHNAN R. (MBT16EE018), ARJUN S. (MBT16EE027), RAHUL S. (MBT16EE085), VISWANTH V. S. (MBT16EE117)** *of VIII Semester towards the partial fulfilment of the requirements for the award of B. Tech Degree in Electrical and Electronics Engineering of the APJ Abdul Kalam Technological University (KTU) during the year* 2020.

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ACKNOWLEDGEMENTS

We thank God Almighty for His Grace upon us during our project work.

First of all, we thank our Project Guide **Ms. Sandhya.P**., Assistant Professor, for providing us all the necessary facilities for the completion of this preliminary project report and also for her timely advice and support.

We also extend our sincere gratitude to the Seminar Coordinators **Dr. Shalu George K**., Assistant Professor and **Ms. Archana.A. N.**, Assistant Professor for their support and guidance for our project.

We also extend our sincere gratitude to **Prof. A S Shajilal,** Head of the Department for his valuable suggestions and guidance for our project.

Last but not the least; we thank our parents and friends for their prayers and moral support throughout the project.

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ABSTRACT

Inter-vehicular networking (IVN) systems have the potential to radically improve the safety, efficiency, and comfort of everyday road travel. Their main advantage is that they bypass the need for expensive infrastructure. In this project we present several major classes of applications and the types of services they require from an underlying network. We then proceed to analyse existing networking protocols in a bottom-up fashion, from the physical to the trans-port layers, as well as security aspects related to IVN systems.

In our project of Inter-vehicular networking it comprises of varied systems including overtaking possibility detection, accident location tracking, automatic braking system and noiseless horn systems which works effectively in reducing accidents and helps in better understanding of road artefact.The Inter- vehicular networking improves performance of the network in terms of transmission reliability, throughput, and interference by introducing cooperation among vehicles. A vehicle-to-vehicle communication protocol for cooperative collision warning, overtaking chances and location sensing. Emerging wireless technologies for vehicle-to-vehicle (V2V) communications are promising to dramatically reduce the number of fatal roadway accidents by providing early warnings.

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CHAPTER 1

1. INTRODUCTION

Annually, 2.3 million people die due to traffic or road related accidents on an average. Without relentless and progressive tackling measures, the cases of traffic accident are hoped to outgrow by 67% by the year of 2022. About 5-11% of the clashes in road/traffic accidents has been caused due to careless overtaking assumptions. To beat this existing difficulty, numerous attempts has been done, one in all of which is the outcome of car, particularly in automobile engineering to branch towards the protection of car.

Automobile networking is one among the foremost essential enabling technologies needed to utilize a multitudinous of applications associated with motorized vehicles, vehicle gridlocks, drivers, riders and jaywalkers. These implications are quite unoriginal and seems a bit far-fetched to see as a goal to a bunch of experimenters and associations. Intelligent Transportation Systems (ITS) aims to focus in the undertaking of vehicles, pull off vehicle pile ups in traffic, aid drivers with safety which is a main cause of concern and other vital subject matters, together with preconditioning the suitability of applications for travelers which are not any longer restrained to testing grounds and research facilities of companies. Fine examples of such works add in to automated toll accumulation systems, driver aiding functions and other vital subject matters encompassing the systems. This integral movement is been fortified by numerous efforts to standardize it and narrows it to form an alliance with other political and industrial bodies whose primary aim is to roll with the leading values, consuming necessaries, and to come up with results for communication methods which mainly involves vehicles and persons within vehicles.

In our project of Inter-vehicular networking it comprises of varied systems including overtaking possibility detection, accident location tracking, automatic braking system and noiseless horn systems which works effectively in reducing accidents and helps in better understanding of road artefact.

1.1 MOTIVATION

The primary motivation towards vehicular communication systems is its safety aspect and also highly reducing the risks regarding unwanted traffic collisions. In keeping

with the World Health Organization (WHO), 1.2 million deaths are caused due to road accidents annually internationally; quarter of all these deaths are caused by accidental injuries. Likewise, about 50 million people suffer first degree or second degree injuries related to traffic accidents. If defensive controls don't seems to be working out, then road related deaths are probably going to be turning into a third prominent reason for deaths during the 2020's from ninth place in 1990's. Nevertheless, the downfall caused by these car scrimmages are in theory unavoidable. This number can be consequently devalued by utilizing local cautionary systems through implying vehicular networking while also reducing blind spots while overtaking. The system also reduces honking at traffic and provides easy user interface between driver and vehicle.

1.2 PROBLEM STATEMENT

Due to the increase in population, the automobile demand also increases. Thus, the causes of accidents are booming day after day. The main cause of such problems are reckless overtaking measures, speeding and poor traffic management. Therefore, providing convenient safety measures to vehicles by incorporating inter-vehicular networking by imposing new requirements for application and services such as, safety messaging, proper and safer overtaking in highways and effective traffic monitoring.

1.3 OBJECTIVES

1. Utmost view of this objective is to cutback the on-road accidents regarding the carelessness of the driver itself.
1. Vehicular networking imposes new requirements for application and services such as, safety messaging, traffic monitoring, better consciousness of road artefact.
2. The system provides service requirements in terms of latency, throughput and reliability on the network.
3. The system provides safety to vehicles as well as the person by sensing, alerting and implementing real time response towards a particular scenario.
4. To decrease the blind spot weakness of vehicles.
5. To provide fast medical attention with instantaneous alertness when an accident occurs.
6. To minimize the excessive noise pollution.

2. ROAD ACCIDENT RUNDOWN IN INDIA

During the global high level Conference on Road Safety held in Brazil in 2015, India has been a scrivener to take the second position assigned to widely as the Brasilia declaration, which, inter-ilia, resolute to cut down by half the deaths and injuries from accidents by 2020 and to reckon this point of reference in the 2030's agenda for sustainable development for the involved countries.Across the 199 countries reported in the World Road Statistics, India ranks 1st in the number of road accident deaths in 2018 followed by China and US. India accounts for almost 11% of the accidents related deaths in the world as per the WHO Global Report on Road Safety 2018. A sum of 4,67,144 road accidents has been reported in numerous States and Union Territories (UTs)during the calendar year 2018, demanding 1,61,417 lives and deeding another 4,69,418 injury cases.

In proportion terms as of 2018, the number of accidents has escalated by 0.47%, severe deaths has raised by 2.47% and casualties have declined by 0.34% over that of the preceding year i.e. 2017. Road accident severity is measured by the number of individual killed per 100 accidents has seen an increment of 0.59% value in 2018 over the former year.

Whilst accidents, deaths and injuries in India attested a sharp rise till about 2010,but the same had reasonably sustained after 2010, with minimal variations from the year to year swings. Furthermore, comparing to the former decades, despite the very high spike in the expansion of automobiles, the compound annual rate of growth on accidents as well as deaths related on accidents in the period 2011-2019 has pull-down dramatically.

Road related accidents are most seminal and is infrequently the result of interaction of diverse factors like (i) human miscalculations (ii) road surroundings and (iii) vehicular stipulation. The content received from the States/UT's on the cause of these road accidents conveys the following:

Over speeding is the leading mistakes, reckoning for 65.1% of the peoples killed followed by driving on the wrong side of the road which encompass about 5.7% of the

accident related deaths as per the category of Traffic Rule Violations. Use of cellular phones computing about 2.3% of the deaths while drunken while driving accounted for about 2.8% of the persons killed.

1. 13% of accidents are caused due to driving without valid license/learner's license.
2. Adding another 29% of deaths can be credited for people not using helmets and another 16% of deaths are contributed by not using seat belts.
3. Vehicles bought more than 10 years accounts for 42% of accident related deaths.
4. Overloading in vehicles accounts for 12% deaths.

In 2017, the State of Tamil Nadu records the highest number of road accidents. Similarly, in 2018 the number of individuals killed in road related accident were higher in Uttar Pradesh. The 54 Indian Urban million plus administrative districts accounts for 18.4% of road related accidents and a 11.6% of road accident deaths in the nation.



Figure -1: ACCIDENT SCENARIO GRAPH

The above graph uncover a conformable hike in road accidents, accident related deaths and road injuries up till 2010 afterward all these segments of accidents, deaths and injuries has been stabilized with light marginal swinging.

3. AUTOMOBILE NETWORKING

Inter-vehicular communication systems are those computer related networks wherein which vehicles act as the communication link, supplying each other with necessary messages like safety related ominous and other vital traffic subject matters. They prove to show a strategic effectiveness in lessening the crashes and foot-dragging. Both nodes are of specific Dedicated Short-Range Communications (DSRC) devices. As a part of Intelligence Transportation Systems (ITS) these vehicular communications has been developed.

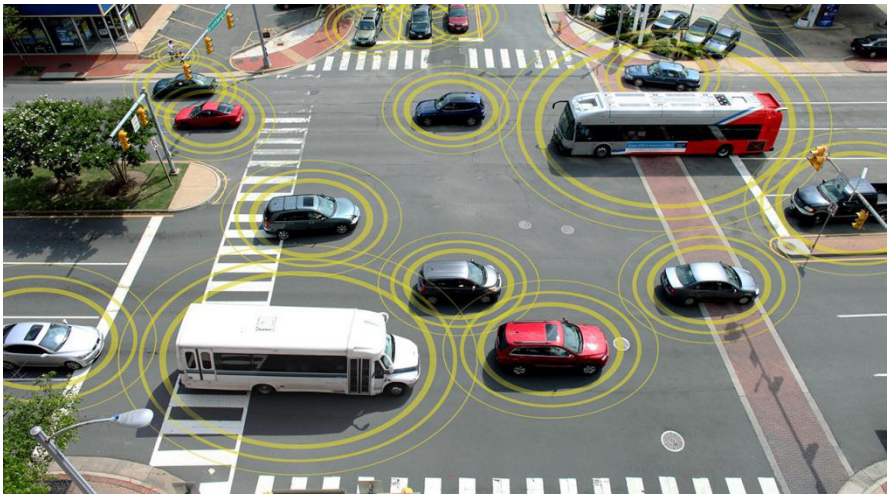


Figure 2: VEHICLE NETWORKING

This type of vehicular networking comprises of various computation and detection technologies for the next phase smart vehicles. Short-range wireless network interfaces, GPS receivers, Embedded computers and potentially access to in-car sensors and also the Internet connectivity are such specialties of these vehicles. Additionally, they are capable of interacting with road-side wireless sensor networks. These abilities are provided into congestion dodging, apportioned computing and sensing operations over vehicular networks for safer driving, in-vehicle amusement or mobile sensing. To influence an oversized scope of such applications, the research designing and execution includes vehicular-specific network code of conduct, middleware programs, and much needed security mechanisms.

3.1 TECHNOLOGY USED

For the system to work, any wireless networking technology is chosen as the base. Considering that the most conspicuous are those short range radio technologies like WLAN (either Wi-Fi or Zigbee). In our project for the sake of simplicity we have used Bluetooth wireless LAN standards as it offers more flexibility than 802.11b but on a smaller range and speed. We have also used another wireless networking called the infrared transmission and reception communication which is used in the noiseless horn implication.

3.2 NEED FOR AUTOMOBILE NETWORKING?

Vehicular networking allows vehicles to extemporaneous and impulsive transfer and acquire SMS from some other cars without needing the help of substructure appliance. As a counterexample, our smart phone relies on the network/infrastructure to access communication channels, connect with other smart phone users, and log on for phone usage. Vehicles can recognize the changing behavior of other cars without wanting the driver's interfacing. Additionally, infrastructure can gain circumstantial consciousness at respective levels of perfectness to come through traffic and possibly enroll towards any preceding emergency services. Modern safety measures, effective vehicle performance and reducing emissions through more consistent speeds are some of the striking advantages as to its necessity.

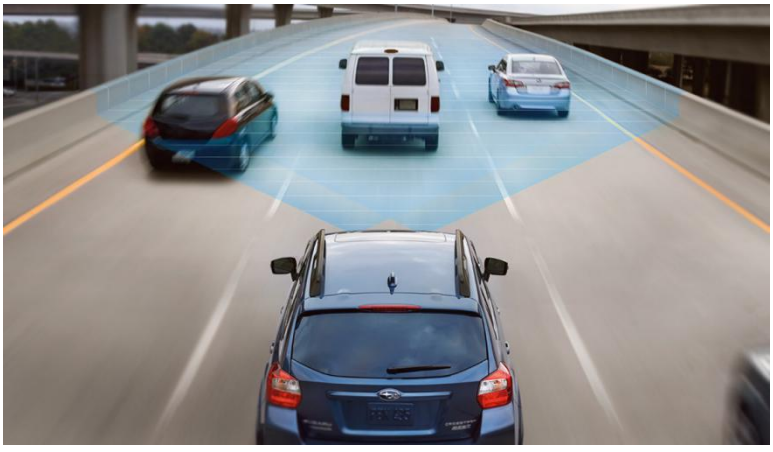


Figure 3: INTERCONNECTING OF VEHICLES

This type of interconnecting vehicles highly reduces the risk of potential accidents while overtaking, speeding and at the time of braking. Various sensors in the vehicles detects the obstacle in front of it and warns the person behind the wheel. Location

tracking and alerting messages can also be sent to others in case of any crashes. Considering the applications intending to why it is used is irrelevant as it is always ahead of the conventional vehicle usage.

4. LITERATURE SURVEY

4.1. Analysis of information dissemination in vehicular ad-hoc networks with application to cooperative vehicle safety systems.

The above quoted paper showcases vehicular collision avoidance support system (VCASS), and demonstration is done by various experimentation with the help of two actual vehicles. Personal computers, telecommunication equipment s, and GPS has been installed inside these vehicles to exchange multiple information and theory ceasing a link. The vehicle could then exchange the location data as well as its rate of change of position with the other vehicles with the help of a UDP (User Diagram Protocol)broadcast and thereby calculates the concerned relative position and speed. Hence these vehicles are able to predict any possibility of collision.

4.2. Development of vehicular-collision avoidance support system by inter-vehicle communications.

The above paper illustrates Cooperative vehicle safety systems (CVSSs) which rely on vehicular ad-hoc networks (VANETs) for the delivery of captious vehicle tracking contents. It uses wireless channels such as WLAN (Wifi/Zigbee) which connects vehicles to a wireless router or nodes and thereby both are within the transmission range to exchange any possible collision warnings. But its main disadvantage is that it struggles to obtain accuracy since vehicle density is changing rapidly at a given time as well as any possible network issue can make the system to fail miserably as it solely depends upon inter-connectivity.

4.3. Overtaking assistant system (OAS) with fuzzy logic method using camera sensor.

This above paper outlines about Overtaking assistant system (OAS) which is one of the Advanced driver-assistance subsystem which is capable of assisting the driver while overtaking and thereby contributes in its decision making. From this experiment,

an OAS is used for overtaking one vehicle on roads whether it is one way or two way. It has been realized with the help of radio control (RC) car. Fuzzy logic is a logic method which uses truth values by considering any real number between 0 and 1(both are included) is employed to handle the idea of partial truth which is used as the overtaking decision-making method. RC car's pixel width is converted into distance between the other car is done with the help of Pixy CMUcam5.Suitable delay will be shown if overtaking is not safe while calculating the fuzzy logic rule.

5. DIFFERENT TYPES OF SYSTEM

5.1. ACCIDENT LOCATION TRACKING

Accident location tracking & Messaging Scheme execution is easy as the system is making use of GSM and GPS technologies. GPS is used with arduino for the purpose of marking the coordinates of the location of the accident while, GSM is used with arduino for the purpose of sending coordinates to the mobile phones. In order to make this system, all controls are activated using the Arduino whereas a LCD is being used for displaying the coordinates. Accident location tracking & Messaging System is very easy and the elements used for this purpose are Vibration Sensor, which recognize the accident and in turn transmits the signals to Arduino. At this period the Arduino gets active and starts to collect the coordinates which are received from the GPS & which are then sent to the Mobile Station using the GSM module. The Vibration Sensor needs 9 Volt of DC supply, if we are choosing a 5 Volt DC power supply, then the circuit will not work correctly. The major units we use here are GSM and GPS modules.

5.1.1. GSM MODULE

Global System for Mobile communication/Global Packet Radio Service module is being used for establishing connection between the computing device and GSM-GPRS system**.** GSM is an architecture which is used for mobile communication in nearly all of the countries**.** GPRS is the extension of GSM which enables increased data transmission rate. GSM/GPRS module has GSM/GPRS modem put together with power supply circuit and connection interfaces (RS-232, USB, etc.) for a computer.

5.1.2. GPS MODULE

GPS is a systematic device in the province of surveillance and tracking, commerce, scientific use. GPS is widely used in all areas except in locations where there will be difficulties for detecting the signal (subterranean location, underwater, inside the building). It is a small electronic device which will permit to link micro-controller board for getting location, height, velocity, date and time on Universal Time Coordinated(UTC). It uses the standardized NMEA protocol for transmitting position data through serial port.

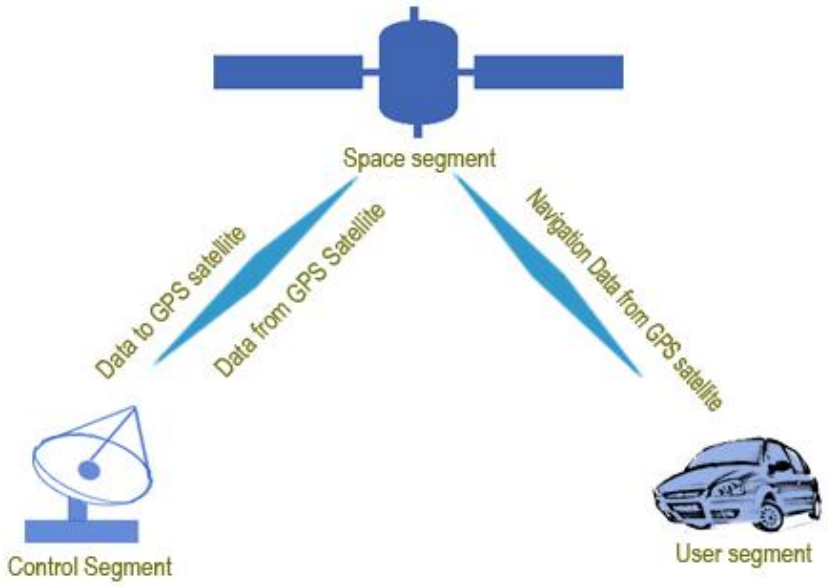


Figure 4: SATELITE COMMUNICATION

5.1.3. 16x2 LCD DISPLAY

A 16×2 LCD display is a base module and it is very frequently used in many devices and circuits. These modules are always chosen over other multi segment LEDs. The reasons are, LCD's are cheap, readily programmable, can display special & even custom (unlike multi segments), animations etc.

A 16×2 LCD shows sixteen characters per line and there will be 2 lines. In this LCD, each chars will be shown in 5×7 pixel matrix. This LCD has 2 registers, which are Command register & Data register.

A command is something like an order given to LCD for doing some tasks like initialize it, spacing its screen, controlling display, setting the cursor etc. The function of the data register id to stock the data that will be displayed on the LCD. The data is something like an ASCII value of the character that has to be displayed on the LCD.

5.2. NOISELESS HORN

This transmits the vehicle horn as an IR signals, rather than audible sound. It uses infrared LEDs and TSOPs for transmitting and receiving IR signals as horn. Buzzers used will produce sound in the inner part of the vehicles for making the driver alert. Normal sound horn is also included with silent horn for making the system congruent with the in use system. The driver can use any horn, which depends on the condition, where in most cases, silent horn can be used.

This selection is done by 2 switches. If the driver is pressing on the switch gently, the noiseless horn will get activated and if more force is applied by the driver, the sound horn will get activated. We are using two toy cars as our prototype, one for housing the transmitter and the other for housing the receiver. This prototype consists of a transmitter & a receiver module

This prototype uses IR LED's as transmitters and TSOP-1738 as receivers. They are IR sensing modules which can sense IR light transmitted at 38 KHz. This device is an active low that can produce bit 0 while receiving infrared light. This output will be connected to 7400 NAND gates, and the output of gates will be connected with LEDs or buzzers. 2N2222 (transistor) is used for the buzzer system. Since the receivers only detect infrared light at 38 kHz, there is a need for generating square wave at the exact frequency. 555 timer IC is used for this purpose. A series of infrared LEDs will be connected to the output of 555 timer for transmitting IR light in all directions.

5.2.1. SOUND HORN SECTION

The normal sound horn switch in the switch section will be connected to the buzzer. This is for the sound horn, which can also be used in most needed situations. This can also be considered as a backup in case, if silent horn doesn't work properly. This section also has Transmitter section & Receiver section.

5.2.2. TRANSMITTER

It consists of a two step switch which is connected to a 12V DC supply. The type of connection between timer IC and buzzer which represents sound horn is in parallel. When the switch for noiseless horn is pressed, current will flow through 555 IC and a square wave will be generated at output. Its frequency will be decided by the resistors (external) & the capacitors. The output will then be connected to the set of Infrared LEDs so that, when they are activated, they will produce IR light at a frequency of 38

kHz. The switch for the sound horn is also mounted on the switch for silent horn. When the switch for sound horn is closed, the connection to the buzzer is completed and a vibration will be produced.

5.2.3. RECEIVER

TSOPs are aligned along 3 sides of the prototype, i.e...left, right and rear. At each side three TSOPs are arranged. Output from all TSOPs are then processed to the comparator for refining the output waveform. Then the comparator output will be fed to NAND gates. Here, two input NAND gates are used. There are two NAND gates in the case of three comparator outputs. Inputs are outputs of first two comparators are given for first NAND gate. The output of third comparator and the output of second comparator is given as the input to the second NAND gate. So, output of second comparator can be given as input for both NAND gates.

5.3. REVERSE AUTOMATIC BRAKING SYSTEM

At present, cars have the alarm system through which, the driver will be alerted when the car comes too close to an obstacle while taking reverse. This will warn the driver and he come to know the obstacle behind the vehicle. But, the above mentioned ideology is likely to have many human error as the driver also is involved in that. Here, this facility is modified by the same system but, instead of an alarm, we are proposing an automatic breaking system where the vehicle can apply brake itself. This system is the advanced version of the same system but we have upgraded it so that the car automatically brakes when an obstruction is close to it based on electronically controlled automobile braking system i.e. “Reverse Automatic Braking System”. The obstacle is detected by an ultrasonic sensor and a control signal will be given to the breaking system. The pneumatic breaking system is used for braking the vehicle. So basically, the vehicle here automatically brakes by determining the distance between the obstacle and the vehicle.



Figure 5: REVERSE BRAKING

An Ultrasonic setup is placed in back of vehicle and it contains of an emitter and a receiving system. The sensor emitter always emits the ultrasonic waves such that whenever a obstacle is detected while in reverse motion, the waves gets reflected and the receiver receives the signal. The reflected waves sends the signal to the Arduino mega micro controller from which it calculates the distance and compares it with the preset value. The arduino measures the distance by calculating the time condemned by the rays to hit the target and reach back such that it is quantified on a mathematical expression : *distance = duration * 0.034/2* . In order for the functioning of automatic breaking, a safe preset value of distance is fed on the arduino such that the calculated distance is always compared. If the distance matches with the preset distance then the brakes are applied automatically along with a message displayed on the LCD panel. If the distance doesn't match with the preset value then the vehicle is good to move backwards without any delay.

5.4. OVERTAKING ASSISTIVE SYSTEM

In normal cases while overtaking, judging an incoming opposing vehicle and trying to overtake another in the same can be risky. This scenario can be nervousness and may even lose control from the vehicle. Thus accidents are of a major cause of concern. But here, the overtaking possibility detection provides assistance while overtaking thereby, showing probability chances of overtaking as well as from the opposing vehicles if any. Thus real time display of the odds are calculated and safety is assured.

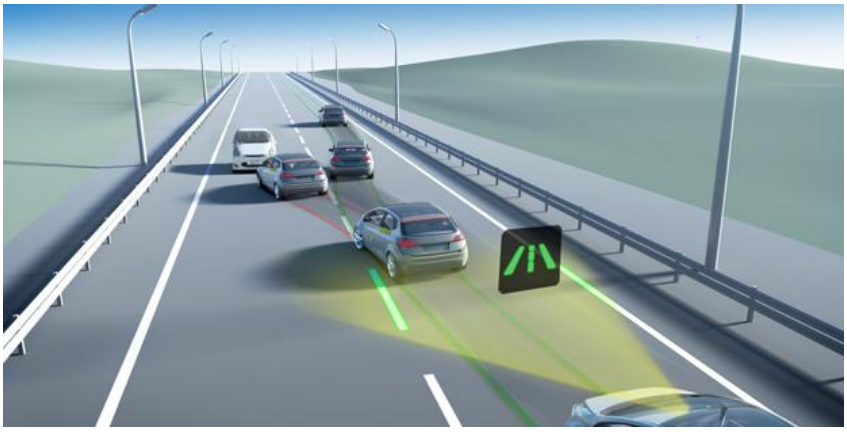


Figure 6: OVERTAKING VEHICLE

For overtaking possibility detection, the ultrasonic sensors used here produces sound waves continuously and checks if, any obstacles are present within a specified range.

The reflected sound waves from the obstacle will be received back by the sensor such that by programming it to calculate the distance, the system could easily find the distance between the obstacle and the source. Arduino is the micro controller used for this purpose. The arduino measures the distance by measuring the time required by the rays to hit the obstacle and reach back such that it is quantified on a mathematical expression : *distance = duration * 0.034/2.* If the distance matches with the preset distance, "OBSTACLE IN FRONT" message will be displayed on the LCD board. If the distance is more than the preset value, then " NO OBSTACLE" message is quoted such that overtaking becomes much safer.

6. SCOPE OF THE PROJECT

To identify areas of increasing human errors while driving and improving not only human and vehicle communication but also with vehicle to vehicle communications and implementing those suitable retrofits to wherever applicable. It also stipulate improvement in safety accompanying method while driving and also enabling vehicles to exchange information regarding safety and traffic analysis. The scope of inter-vehicular networking application has accumulated with the recent progress in technology and improvement of astute cities across the globe.

7. BACKGROUND

Mar Baselios College of Engineering and Technology was established on 2002, at Bethany hills Trivandrum aimed at obtaining a detailed idea about various ad hoc networks and its communications with each other thereby imparting this idea towards narrowing to inter-vehicular communication. The target is to achieve personal safety and comfort while driving.

CHAPTER-2

1. INTRODUCTION

Modern survey show that about sixty percentage of roadway accidents could be obviated if the driver was informed just one-half second before the accident occurs. In reality, traffic accidents have become the main reason of mortality, quite above malady. Future technologies appears to provide quicker, safer and more dependable communication techniques. Draw together, this communication can be used systematic to reduce accidents, as well as to assist and improve the attribute of the traffic. There are various kinds of systems to maneuver drivers in the roads. This paper is centered on four systems that are overtaking possibility chances, accident location tracking, automatic reverse braking system and noiseless horn. The main goal is to raise on-board devices (i.e. GPS, sensors) and, thus, to widen the range of drivers. Overtaking maneuver is one of the most risky scenarios for road vehicles particularly in two-way roads. The overtaking plan of action, which is a driver assist function or rather an autonomous function in electric/autonomous vehicles. The method uses sensor and signals from the vehicle to the surroundings. These signals are processed with cluster methods in order to attain probability density functions and anticipate their predicted movement. The vehicle detection mechanism which automatically SMS to personal contacts with the information of vehicle location when an accident happen using the GSM/GPRS system. The SMS send to different regimen contains the info of the vehicle and its location. If any firing take place in the vehicle, then the fire sensor sends message to the driver's selected contacts. Hence with this project implementation we can detect the position of the vehicle where the accident has happen so that we can provide the first aid as early as required. A smart backward braking method is compiled with ultrasonic sensor which performs a pneumatic braking system. The main reason of this system is cars can endeavor backward automatic braking system when the sensor senses the obstruction. The 'Noiseless Horn System' is an innovation in the present car horn system. It uses infrared LEDs and TSOPs to transfer and perceive IR signals as horn. Buzzers are provided to create sound inside cars to alert the driver

1. BLOCK DIAGRAM

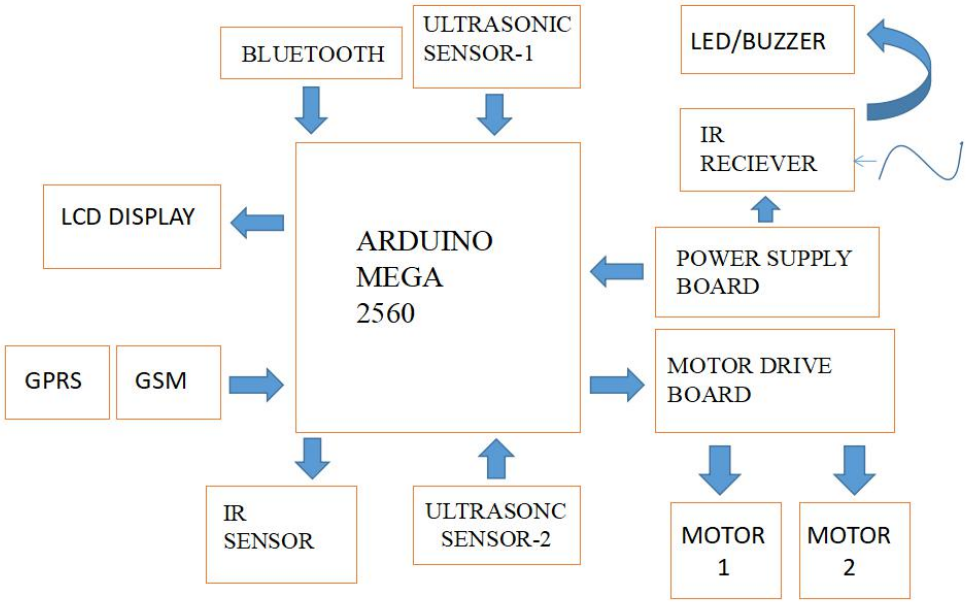


Figure 7: BASIC BLOCK DIAGRAM.

1.1. BLOCK DIAGRAM DESCRIPTION.

2.1.1. Bluetooth -The model of the vehicle is controlled via blue-tooth connection with the help of a mobile phone app called blue stick control which provides the signal for forward and backward motion of the car. The blue-tooth module is attached to the arduino mega board for processing the specified function

2.1.2. Ultrasonic Sensor 1- This ultrasonic sensor is put-upon to find the overtaking possibility chances in which it emits a sound wave which deflects from the obstacle ahead and this deflected signal is received through the receiver with set of information which is processed by the arduino for displaying the chances in an LCD display.

2.1.3. Ultrasonic Sensor 2- This ultrasonic sensor is used for reverse braking system which provides sound wave to deflect from an obstacle at the rear side of the car. While reversing this ultrasonic sensor continuously transmits and receives the wave and measures the distance between the vehicle. Whenever the obstacle is at a distance of limited value which is set in the arduino, it provides a stopping command.

2.1.4. Motor drive board- The motor drive board consist of pins C1-A, C1-B, C2-A, C2-B. These pins are directly connected to the motor1 and motor2 respectively which

provides the forward, backward, left, right motion of the vehicle. This board is also electrically connected to arduino mega board which helps us to control motor drive board via blue-tooth for their movements.

2.1.5. Power supply board- This board provides the input Power supply to each and every module from an external source. This board consist of 3 sets of pins i.e. 12v, 5v, and ground. The arduino, LCD display, motor drive board, IR sensors and two ultrasonic sensor requires a 5v supply and for the blue-tooth module a3v supply is required. Each every supply connection can be provided with a single power supply module

2.1.6. IR Sensors (Transmitter and Receiver) - An IR sensor also known as infrared (IR) sensor is an electronic instrumentation that measures and detects infrared radiation in its surrounding environment. The IR sensors are provided for noiseless horn in which the sensor transmits an IR signal to alert the nearby vehicle which receives the signal through a receiver. A pair of both transmitter and receiver is placed in each car to make alert as well as to receive alert.

2.1.7. Arduino mega 2560 - It is the head part of the arrangement which provides all the commanding and processing signal to various modules to perform their functions. It consists of 53 pins. All of the fifty four digital pins on the Arduino 2560 Mega is used as an input as well as output, using digitalWrite(), digitalRead(), pinMode(), and functions. They run at five volts. Each pin can supply or acquire a peak of 40 mA and has an interior pull-up resistor of 20-50 kOhms. Since it contains more than one system, we required arduino mega board instead of adruino uno board to occupy all the pins.

2.1.8. LCD Display- LCD (Liquid Crystal Display) screen is an electronic device which is used as display module and find a wide tract of applications. A 16x2 LCD display is very fundamental device and is very normally used in different equipment and circuits. These modules are preferable over 7 sections and other multi sections. The LCD Display helps to show the information regarding the overtaking possibility chances and also show the distance between the obstacle from the car.

3. WORKING

Inter-vehicular networking is a combination of different sub-systems in which each system has different working methodologies and functions. The four systems focused on this project are overtaking possibility chances, accident location tracking, automatic reverse braking system and noiseless horn. Therefore, working of each system are as follows.

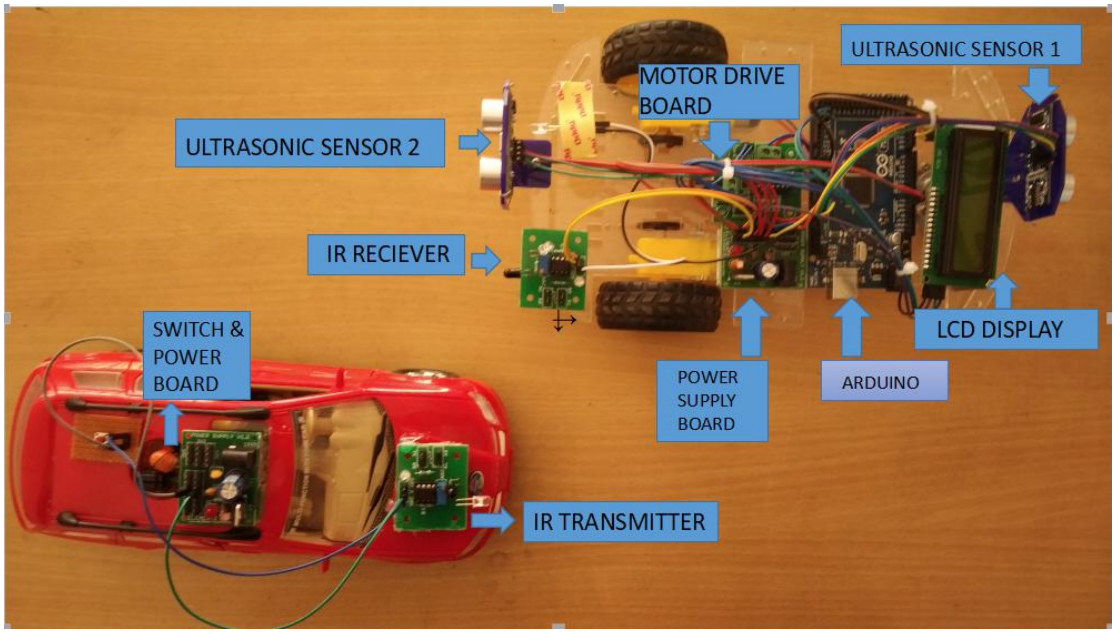
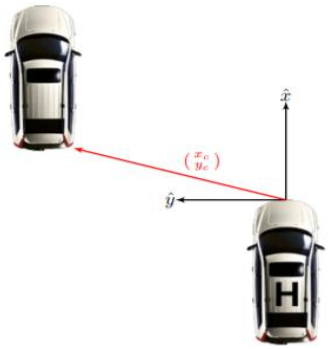


Figure 8: PROTOTYPE OF THE VEHICLE

3.1 WORKING OF OVERTAKING ASSISTANCE SYSTEM.

The working of overtaking assistance system is used with the aid of ultrasonic sensor which is mounted on front and sides of the car. The ultrasonic sensor is a sensor which contains two pin trig pin and echo pin in which both pins are simultaneously provided with low, high and low input pulse which generates a sound wave and is transmitted by the transmitter and this transmitted wave is deflected if an obstacle is in its path. Ultrasonic sensors works by producing a sound wave at a frequency more than that of human hearing range. The transducer of the ultra sensor perform as a microphone to perceive and transfer the ultrasonic sound. The ultrasonic sensors use an individual transducer to send a wave and to perceive the echo. The sensor determines the spacing to a target by measuring time lapses



between the sending and perceiving of the ultrasonic pulse. Suppose the car is overtaking another car, the ultrasonic sensor continuously emits this sound wave and if an obstacle is detected this sound wave reflect back and received by the receiving system of the sensor thus by determining the actual time difference that is required by the sound wave to receive back. By simple algorithm of speed and time the actual distance from the car and the obstruction is determined. This ultrasonic sensor is electrically connected to arduino mega 2560 board which contains algorithm to show the information on the LCD display. The arduino is preset-ed with min value of distance that is required to successfully overtake a vehicle and if the value become less than the min value then “obstacle in front” is shown in the display to alert the driver.

3.2 WORKING OF AUTOMATIC REVERSE BRAKING SYSTEM.

The working of automatic reverse braking system is also used with the help of ultrasonic sensor which is placed at rear side of the car. The ultrasonic sensor is a sensor which contains two pin trig pin and echo pin in which both pins are simultaneously provided with low, high and low input pulse which generates a sound wave and is transmitted by the transmitter and this transmitted wave is deflected if an obstacle is in its path. Ultrasonic sensors works by producing a sound wave at a frequency more than that of human hearing range. The transducer of the ultra sensor perform as a microphone to perceive and transfer the ultrasonic sound. The ultrasonic sensors use an individual transducer to send a wave and to perceive the echo. The sensor determines the spacing to a target by measuring time lapses between the sending and perceiving of the ultrasonic pulse.

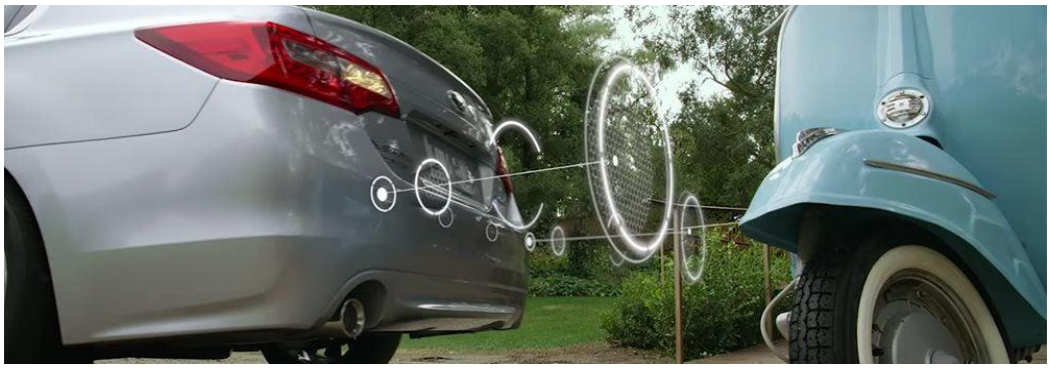


Figure 9: REVERSE BRAKING SYSTEM

The ultrasonic sensor continuously emits this sound wave and if an obstacle is detected this sound wave reflect back and received by the recipient of the sensor thus by determining the actual time difference that is required by the sound wave to receive back. By simple algorithm of speed and time the actual distance from the car and the obstruction is determined. The arduino is preset-ed with min value of distance that is needed to stop the vehicle. Whenever the car reaches its limited value of distance between the obstacle the arduino initiates the stopping command to the car.

3.3 WORKING OF NOISELESS HORN SYSTEM

The noise less horn system is used to reduce the surrounding noise pollution with the help of IR sensor which contains a transmitter and receiver. Therefore, it consists of a transmitting end and receiving end. The transmitting end has a 12V DC source is attached to the double press switch. Collateral link is made up to the buzzer and timer IC to act as the sound horn. When the button for soundless horn is pressed,

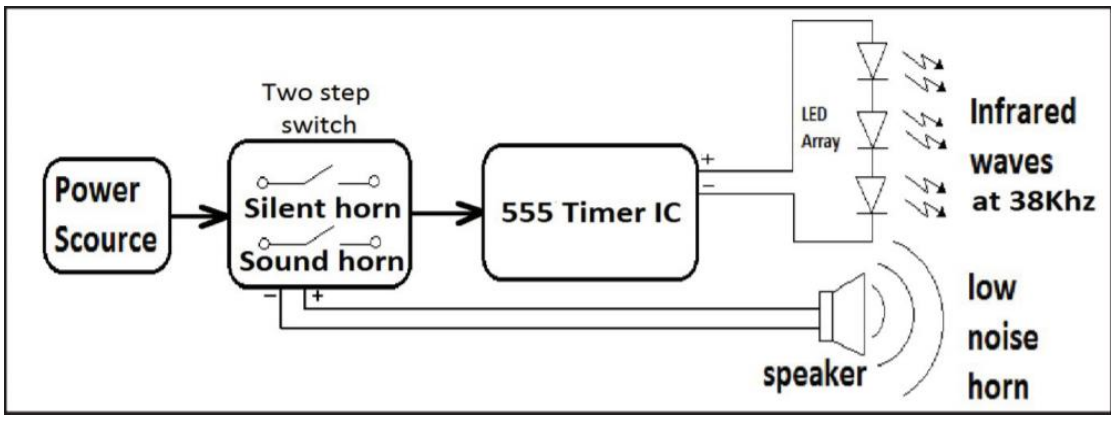


Figure 10: TRANSMITTER CIRCUIT

Current will flow from 555 IC and a square wave is produced at the output, which has adefinte frequency by the outside capacitors and the resistors. The corresponding output is attached to a pack of infrared LEDs, so that whenever they are excited, they produce IR light at a frequency of 38kHz. The normal horn button is also affixed with the noiseless horn switch. When the normal horn button is closed, the connection to the buzzer is consummated and a buzzing sound inside the vehicle is obtained. The receiving end has TSOP sensors are ordered on three sides of the vehicle, mainly, back right and left. On each side there are three sensors. Output of each sensor is granted to comparator to amend the output waveform. For the project, we use 2 cars, one for lodging the transmitter and the other for lodging the receiving system. This

work employ TSOP1738 as receiving system and simple IR LED as transmitters. TSOP1738 are IR sensors, that signifies IR light modulated at 38Khz. It is an active low maneuver, and obtains bit 0 on perceiving Infrared light. This output is attached to 7400 NAND gates, and the output LEDs/Buzzers are attached to the output of gates.

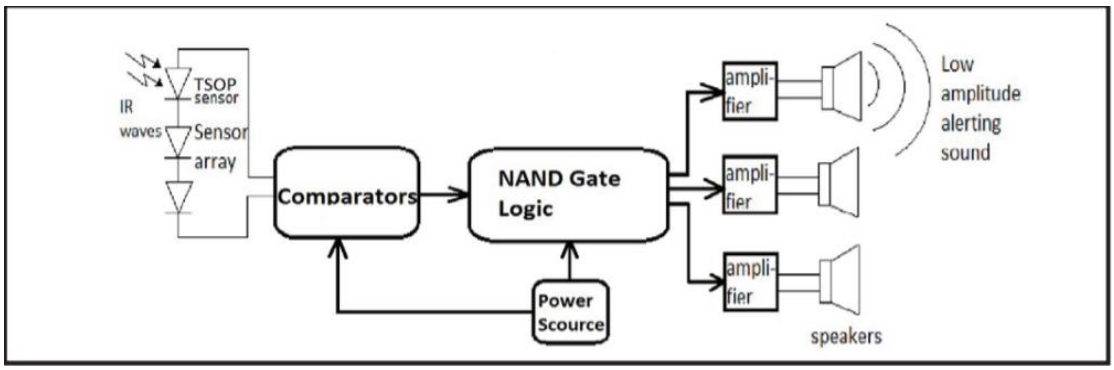


Figure 11: RECEIVER CIRCUIT

3.4. WORKING OF ACCIDENT LOCATION TRACKING.

GPS based accident location spotting system for ambulance includes an accident locator instrumentation and SMS scheme. A hardware model was developed using constituent like GPS and GSM components. An algorithm is used which alter better efficiency in inquiring the closest hospital to the accident area. The function of this system will intensify the current exigency response service by render a platform for coverage and responding to accidents expeditiously.

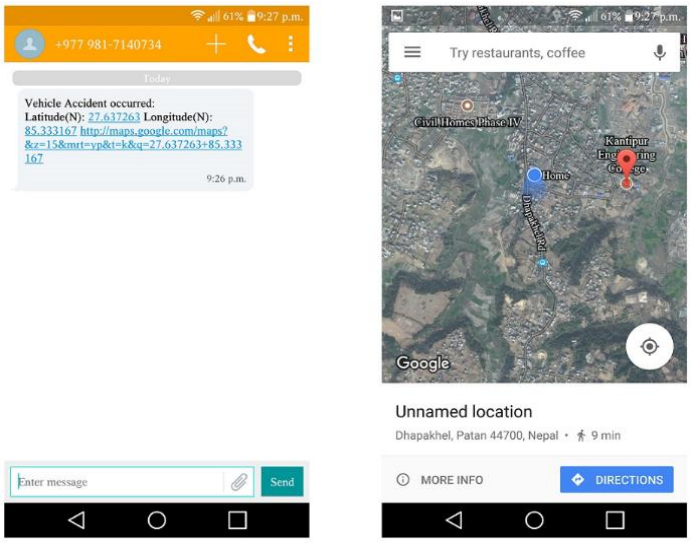


Figure12: MESSAGING AND LOCATING

This will alter the authorities to render faster medical aid to the public, which will decrease fatality rate. The Mercury switch used in this work as an input sensor which is to discover the accident impact. Whenever a vehicle strike an obstruction the collision is perceived by the mercury switch and it transfer signal to the micro controller. The GPS module is used to get the position co-ordinates, it instantly perceive the signal from satellite. GPS receiver perceive message signals from the satellites and measures its distance from satellites. It is done by calculating the time needed for the signal to travel from satellite to the receiving system. Where, Speed = Speed of signal which is roughly equal to the speed of light Time = Time needed for a signal to travel from the satellite to the receiving system. By deducting the transmitted time from the received time, we can find out the travel time to determine the distance, both the satellite and GPS receiver create the same pseudo code signal simultaneously. The satellite transfer the pseudo code; which is received by the GPS receiving system. These 2 signals are analyzed and the variation between the signals is the total time of travel. Now, if the receiver see the distance from three or more satellites and their localization, then it can measures its position by using Trilateration method. To raise the GPS receiver a easy command GPGGA is used. The GSM module is used to communicate with the micro controller to send messages to smart phones through UART. To communicate over USART or UART, we require 3 fundamental signals which are namely, TXD (transmit), GND (common ground), RXD (receive). The GSM modem which is attached to PC is intersection through optic basics software used to design an application to supervise the accident position of the vehicle.

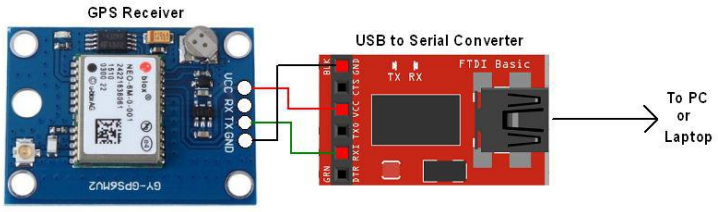


Figure 13: GSM AND GPS MODULE.

Trailing the position of the car even in Stygian clunky places where there is no network for perceiving the signals. In our project GPS is used for trailing the location of the vehicle, GSM is provided for transmitting the SMS. Hence with this project execution we can find the location of the vehicle immediately after an accident occurred so that we can stipulate the first aid as fast as possible. The project presents

vehicle accident spotting and vigilant system which message to the user distinct phone numbers. The GPS trailing and GSM alert-based algorithmic program is organized and enforced. Therefore this accident location tracking system can trail geographical data automatically and sends an alert message about the accident.

4. COMPONENTS

4.1 ARDUINO MEGA 2560

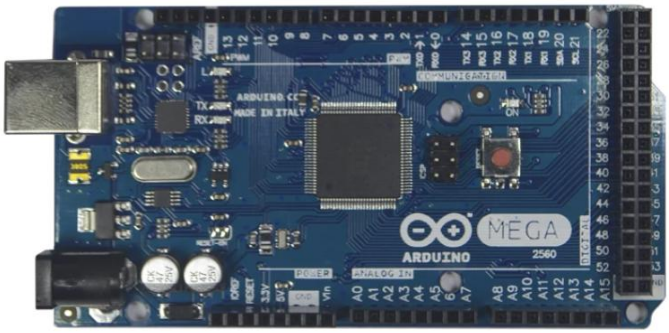


Figure 14: ARDUINO MEGA 2560

It is a micro-controller based board supporting this Atmega2560. It has numerous numbers of memory pins and I/O pins when looking with other boards. There are 4 digital input-output pins and it also has another 16 analog pins that makes this device different from others devices. Out of fifty four digital Input/output pins, 15 pins are used for PWM.

This controller contains of quartz oscillator of 16MHz and has a port for USB cable to connect and convey various codings from computer to controller.A DC powered connector is used to cater power to the board. It has two voltage regulators; 5V and 3.3V which renders the credibility in managing the power. It also has a large memory, size and many number of I/O pins

4.1.1. PIN DESCRIPTION

1) **5V & 3.3V** It provides output regulated supply voltage to about 5volts. This regulated power supply is capable of providing power to board and other units on the board. It's been attained from Vin of the controller or USB cable or some other regulated 5volts supply while, 3.3V pin provides another power supply boar pins. 50mA is the maximum power it can draw.

2) **GND**. There are 5 ground pins present on the board for connections.
3) **Reset**. This pin is used for when a case arise in resetting the board. Place it LOW for resetting the board.
4) **Vin.** This is the input voltage to the board and it ranges from 7volts to 20volts. Power connector provides the required potential difference that can be provided via this pin. No matter what the output voltage supplied, the board automatically sets it up to 5volts, and the power connector gives the required voltage.
5) **Serial Communication.** Pins used to transmit and receive serial datas here are RXD and TXD i.e. the transportation of data is represented by (RX) while, (TX) is used for receiving data. 4 sequence of these serial pins has been used where Serial 0 consists of RX(0) and TX(1), Serial 1 consists of TX(18) and RX(19), Serial 2 consists of TX(16) and RX(17), and Serial 3 consists of TX(14) and RX(15).
6) **External Interrupts**. Total 6 pins are used for creating interrupts (external). They are interrupt 0(0), interrupt 1(3), interrupt 2(21), interrupt 3(20), interrupt 4(19), interrupt 5(18). These pins supplies interrupts by numerous ways i.e. granting LOW value/rising or falling edge /fluctuating value to the interrupt pins.
7) **LED**. The controller has an in-built led connection for the digital pin 13. when it is set to HIGH value the pin turns the led ON and LOW value turns it OFF.
8) **Analog Refrence voltage(AREF).** AREF (Analog Reference Voltage) is a reference voltage for the given analog inputs.
9) **Analog Pins.** These sixteen analog pins are embedded within the board which are tagged as A0-A15. However, these analog pins can also be used as digital Input/Output pins. All analog ports have 10-bit resolution. Nonetheless, the upper value can be transposed using AREF and analog Reference () function.
10) **I2C.** Two pins 20 and 21 supports I2C communication wherein the 20 shows SDA (Serial Data Line mainly used for holding the data) and 21 shows SCL (Serial Clock Line preferably used for subjecting data synchronization amidst devices)
11) **SPI Communication.** SPI stands for Serial Peripheral Interface and is used for the transmitting datas between controller and other peripherals. 4 pins i.e. 50 (MISO), 51 (MOSI), 52 (SCK), 53 (SS) are used in SPI communication.

4.2. ULTRASONIC SENSOR

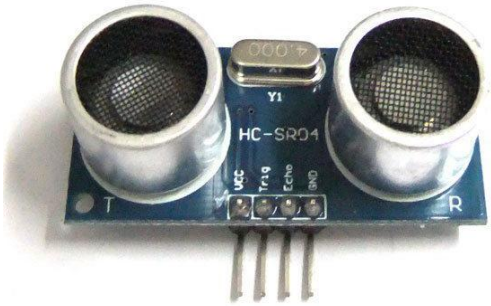


Figure 15: ULTRASONIC SENSOR

An ultrasonic sensor evaluates the displacement of an object with the aid of ultrasonic sound waves. It has a transducer which can transmit and receive these sound waves that is capable of providing the information about any possible obstacle nearby. High-frequency sound waves reflects from any blokage to supply various echo patterns. The working methodology is similar to that of a radar system.

An ultrasonic sensor converts electrical energy into waves and vice versa which can travel at a frequency above 18 kHz. A micro controller is used to process the information of the ultrasonic sensor. Whilst measuring the distance, the micro controller gives an initial trigger signal to the ultrasonic sensor. When triggered, the ultrasonic sensor will generate 8 acoustic wave bursts and develops a time counter. Suddenly when the reflected (echo) signal gets received, the timer stops. The equation for tabulating the distance between the sensor and obstacle in front is,

$$\text{Distance} = \frac{\text{time taken} \times \text{speed of sound}}{2}$$

Equation 1

Theoretically, the distance can be calculated using the Time/Rate/Distance measurement formula. Since the measured distance is the distance traveled from the ultrasonic transducer to the object and received back to the transducer it is a two-way trip. By dividing this distance by 2, you can find out the actual distance from the transducer to the object. The speed of Ultrasonic waves can be taken as the speed of

sound i.e. (343 m/s). The micro controller interpret the time signal into distance using this equation.

$$\text{Distance(cm)} = \frac{\text{echo pulse width } (\mu\text{S})}{58}$$

Equation 2

4.3. LCD DISPLAY



Figure 16: LCD DISPLAY

It is an electronic display that has a wide variety of applications. A 16x2 LCD display is the basic system which isordinarily used in devices. The reason is, LCD's are cheap, readily programmable, can exhibit special & moreover custom (conflicting multi segments), animations etc.

A 16×2 LCD shows sixteen charas each line and it has 2 lines.Here each character is displayed in 5×7 pixel per intensity matrix.It also has 2 registers, namely Command & Data registers.

A command is nothing but an order given to LCD for doing some tasks like determining it, cutting off its screen, scoping the cursor, predominating display etc. The function of the data register is to store the data and to be displayed on the LCD. The ASCII value of the character is the data and it has to be displayed on the LCD. The purpose of its usage in our project is to display the exact distance of the system and to show whether the path is clear or not to help the driver take decision accordingly. In this LCD display we are also able to show the exact distance while reversing as it shows the distance from the obstacle at back side of the vehicle while reversing. It can display exactly 16 characters horizontally and has a capacity to fill in two rows of sixteen characters.

4.4. IR SENSORS.



Figure 17: IR SENSOR

It is an electronic device that is capable of measuring and detecting IR radiation in the possible environment. IR is invisible to the human eye, as its orientation is longer than that of the visible light. IR sensing elements are little microchips with photocells that are tuned to comprehend the infrared light.

The TSOP sensor is a type of IR sensor that has the quality to interpret the output signals like as in television remotes,Home theatre remotes, AC remotes etc. Each and every remote will work within a frequency range of 38kHz and the IC can construe any IR signals and thereby processing them by providing output to pin 3. TSOP-1738 acquires only 38Khz Infrared signals.TSOP-1738 is an Infrared Receiver Sensing element, which can be used to receive Infrared signals of 38Khz. The sensor run on 5V and takes up around 5mA to function. Normally the signal pin (pin 3) IC is connected to a micro controller to examine the IR signal received. The IR transmitter comprises of an LED emitting the IR radiation. This is accepted by the photo diode, which acts as the IR receiver at the perceiving end. IR radiation is perfect for using in wireless communication as it is invisible to the human eye.The electronic remote primarily consists of IR transmitter and receiver units. A remote control showcases a flash of non-visual light which turns into any instruction and is accepted by the receiver unit. This component is used for noiseless horn system in which we use IR rays to communicate between the drivers instead of the normal horn to minimize the excessive sound pollution

4.5. GPS MODULE.



Figure 18: GPS MODULE

The sensor a satellite-based system which is abbreviated as Global Positioning System, that make use of the satellites and ground stations locate its position on Earth. GPS receivers are broadly employed in smart phones, traffic governing systems, in defence etc. GPS receivers receive packets from 4 satellites for maintaining correct precision and accuracy. GPS receiver doesn't transmit any data signals to the satellites. These GPS satellites transfer data signals via frequency (1.1 - 1.5 GHz) to the receiving system. With the aid of this specific data, a GPS module will come to know its position and time. Further, it reckons the distance from these satellites. This can be done through by taking the time taken by the signal for travelling from satellite to the receiving system. By taking off this time from the collected time, its been able to confirm the period of time to work out distance, such that the satellite and GPS receiving system renders a similar fake code signal on a similar time. The satellite carries the fake code; that is accepted by the GPS receiving system. Both these signals are filtered upon and the dicrement between these signals is the time period . Instantly,the receiving system is aware of the gap from three or additional satellites and their placements (which are vesseled by the satellites), then it will calculate the exact location using Trilateration method. NEO-6M is an international positioning system (GPS) module, a really widespread, efficient, superior GPS module with a ceramic patch antenna, associate on-board silicon chip, and a backup battery that may be conveniently integrated with a variety of small specified controllers.

4.6. GSM MODULE

The GSM modem used is SIM 900 and it mainly consists of 3 connections, RX, TX and the ground. The maximum voltage that can be given to the GSM modem is 12V. It consists of 3 LEDs.ie, the network LED which indicates whether the range is available or not. The board LED indicates whether the board is in proper working condition or not. The ON and OFF LED indicates whether the modem is on or off. Serial connection is usually used. If it is not available, cross connection of RX and TX can be made.



Figure 19: GSM MODULE.

For our project of connecting a GSM to arduino and therefore to send and receive SMS by processing via an arduino is through TTL Output provisions. SIM900 GSM Module means that the module is capable of supporting communication at 900MHz band. Each of them have completely different input power offer specs. GSM module needs a twelve volts input. Such that, we tend to provide it by employing a 12V,1A DC. In whatever case, the communication between arduino and GSM module becomess serial it is quite compelling to use the serial pins of arduino (RX and TX).

4.7. BLUETOOTH MODULE

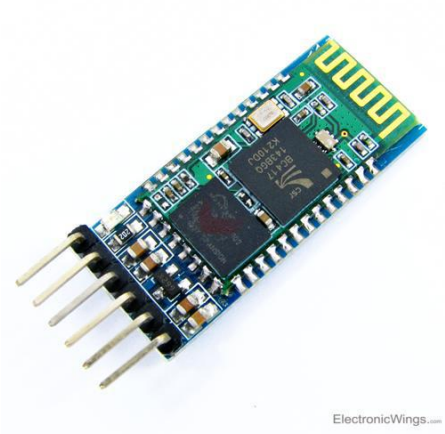


Figure 20: BLUE-TOOTH MODULE.

HC-05 is the bluetooth module used here.The bluetooth devices act as the master and the slave. Bluetooth serial compartment allows all sequential enabling devices to intercommunicate with each other using using this. It has several applications some of them constitutes to wireless earphones, gaming controllers, wireless speakers, wireless mouses and plenty of such varied applications.Ranging can be up to <100m which depends upon the sender and recipient, ambiant conditions. It has 6 pins which are discussed below,

1.Key/EN: Its main purpose is to bring the bluetooth module towards the AT command mode. If Key/EN pin is placed high, then the module will be in a command mode.Nonetheless by default it will be in data mode. 38400bps is the default band rate of HC-05 in command mode is and 9600 bps is its data mode. The module also has two modes,

1.1. Data mode: This mode is set when exchange of data is required or changing of certain information inbetween the devices.

1.2. Command mode: AT commands are used to change the required settings of HC-05 module. Serial (USART) port is used to send certain commands to the module.

2.VCC: pin is used to connect a 5volts or 3.3volts to the module.

3.GND: This pin is used for provideing ground configurations to the module.

4.TXD: Transmit Serial data (The data will be sent out serially on TXD pin by Bluetooth module)

5.RXD: Receive data serially (to acquire the data wirelessly by the bluetooth module)

6.State: It tells the user whether the bluetooth module is connected or not.

A red LED shows whether the main connections in the Bluetooth are the ground CC, RX and TX. The Bluetooth and Arduino are cross connected. The connection between the Arduino and the Bluetooth can be either software serial or the hardware serial. RX and TX connection should be specified if software serial is used. Cross connection is made when hardware serial is used.

4.8. MOTOR DRIVE BOARD

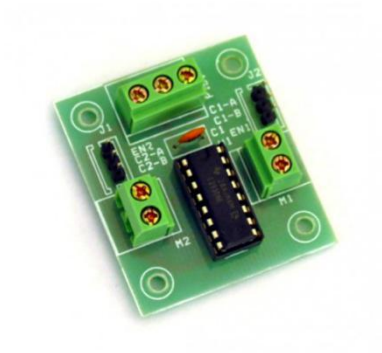


Figure 21: MOTOR DRIVE BOARD

L293D shield may be the driver board supporting L293 IC, which are capable of driving the four DC motors and a couple of stepper motors for an equivalent time. From each one channel of this module it has an utmost current of 1.2A and it wil not work if the voltage is over by 4.5volts. And then, its a task on a lookout for picking out the proper motor with its nominative voltage and current. L293DD has 4 outputs which makes it suitable for 4-wire stepper motors. L293D also can be wont to drive servo motors. There are some remarkable reasons for selecting L293D, thos are the low-cost (compared with other actuations), seemless shape and sizing, ease to control, no needing of any cautionary circuits and diodes, no needing for any such warm sinks and decent resistance to temperatures as well as towards high-speed stochastic variables.It is readily found in motors having a voltage between 5volts to 36volts and the current scaling of upto to 600 mA. Nevertheless, itha sthe capability to withstand a current up to 1200 milliamperes in 100 ms. The frequency of the IC is 5 kilohertz. We use this board for the forward, backward, right and left motion of our vehicle. This drive board is electrically connected to arduino mega board which is programmed to drive the

motor consistent with the users will. It consists of a 5V, ground and 12V. These pins are used to give connections to the board. It consists of two set of pins: C1A, C1B and C2A, C2B. These pins are used to control the movement of the wheel and it is also used to correct the direction of the motor.



Figure 22: MOTOR

The Two Battery Operated DC series motors are used for the movement of the vehicle.

- Operating voltage 3-12 V
- Rated speed 60 rpm
- Rated torque 1 kg-cm

4.9. POWER SUPPLY BOARD

This is a DC-DC 12V to 3.3volts-5volts-12volts module having a multi-purpose voltage conversion which is perpetually called as the buck converter/step-down voltage converter. The module is capable of neutering the output of the power source before supplying it to the load such that it becomes much safer on delivering the current.

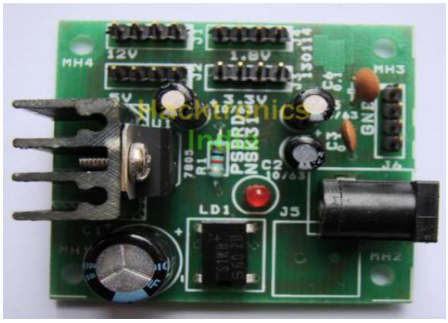


Figure 23: POWER SUPPLY BOARD

The device is extremely seemless and straightforward for typical usage. The module is hopped-up by 6volts to 12volts DC input, and stipulates to the three fixed DC outputs: 3.3V, 5.0V, and a third output signal which might be an immediate connection to the DC input. They can be used as a multi-purpose fixed voltage DC power supply in other instrumentation designing. So larger the duty cycle,superior the output voltage from the power supply. The female barrel jack connector on the board is the input terminal. Terminal blocks on the board enable the connection of components using the male bread board wires.Voltage regulation is acquired by changing the duty cycle with high sped switching up to 200kHz.Smaller components are often used so as to reduce the size and weight of switch mode power supply.

4.10. ACCELROMETER



Figure 24: ACCELEROMETER

Acceleration is the rate of increase of speed of an object such that inorder for its calculation an accelerometer is used The implication of accelerometer diversifies to vast directions, such as in laptops, it shields the arduous drives from any careless falls. Accelerometer are besides employed in automobiles because of the emerging trade methodology for the detection of any possible car crashes and deploying the airbags all of a sudden. In our project the accelerometer plays a role in sensing any external forces implying to the vehicle inorder to give the information to the arduino. Arduino then quantifies this piece of information to the threshold value and if it is higher than that, it then arrives into a conclusion that the car met with an accident and triggers the gsm and gps modules to send the accurate position and location to the already

linked mobile number. If external force is below rhe threshold value it will not contact the processing unit.

5. RESULT

The following results were obtained by integrating the different components from testing the system under given conditions.

1. Noiseless horn section of the system was completed by integrating all the components required with ardiuno and programming was done successfully.

2. Motor driver board was implemented with ardiuno and programming was done for a mobile device to control the movement of the vehicle using a Bluetooth module.

3. Ultrasonic sensor was programmed with ardiuno with a pre-set value of 30cm for the prototype for overtaking possibility detection where, the system successfully provided a warning message when the obstacle is less than 30cm and displayed the message on the LCD display.

4. Ultrasonic sensor was programmed with ardiuno with a pre-set value of 20cm where, the system successfully stopped in front of an obstacle 20cm away from it.

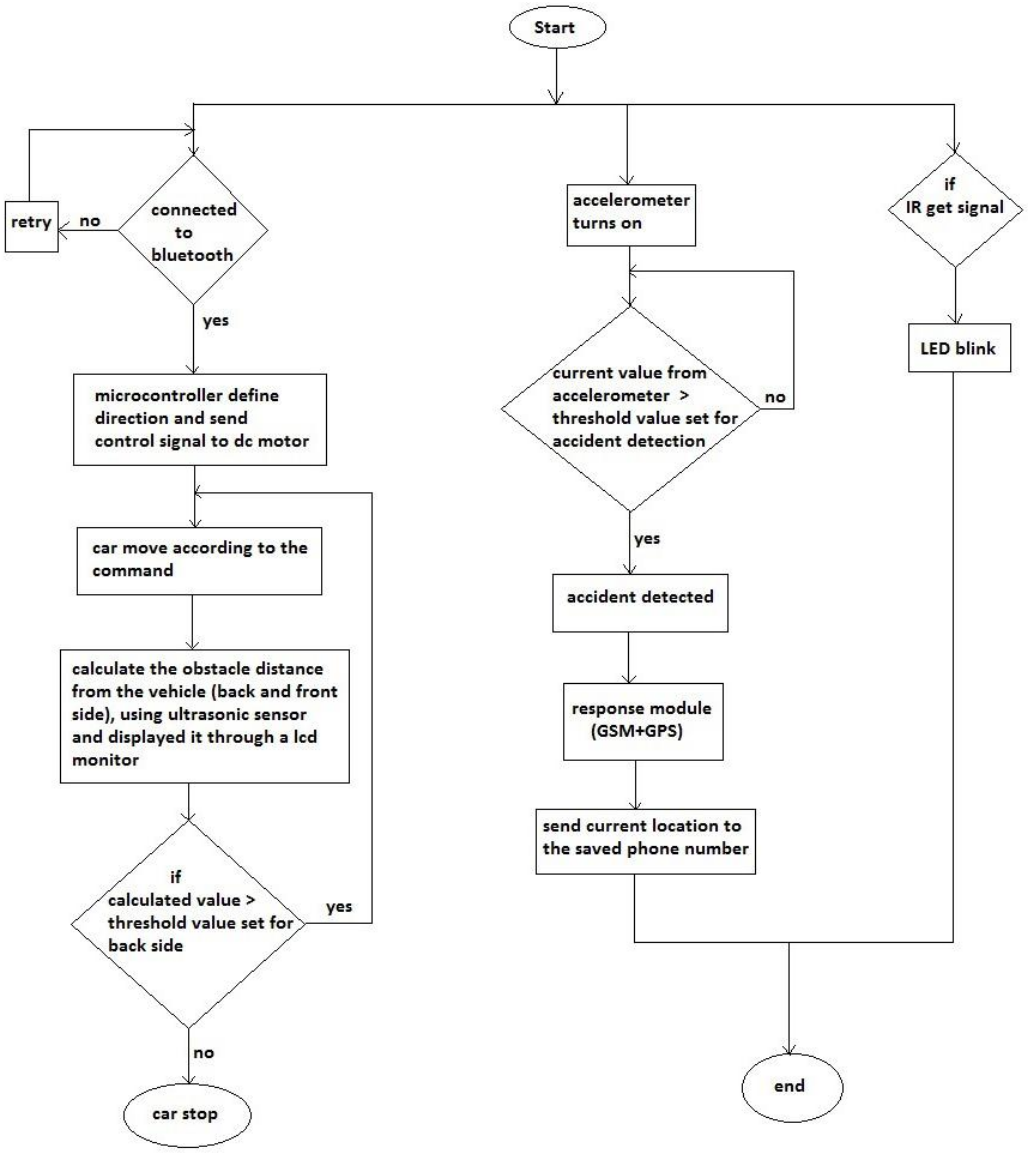
6. FUTURE SCOPE

Due to rapid increase in modern technologies and the automobiles using this technology for better experience in driving the safety of the passengers are also essential to be considered. In future automobile networking will become an important part of our life which we depend to have smooth and easy experience.

The idea of this project is the reduction of the road accidents which are increasing day by day mainly due to the carelessness of humans. By implementing these systems, the vehicle could definitely assist the driver in taking some decisions which will be very useful for the inexperienced ones.

In case if an accident occurs, the location tracking system will instantly provide the message to our close ones so that, they could offer some help. We can easily see the reduction in the road accidents after the implementation of these systems and can be taken for further studies. Noiseless horn in the system aids in reducing the noise pollution and the driver will only give a sound horn if needed.

7. FLOWCHART


Start
retry
no
connected to bluetooth
yes
microcontroller define direction and send control signal to dc motor
car move according to the command
calculate the obstacle distance from the vehicle (back and front side), using ultrasonic sensor and displayed it through a lcd monitor
if calculated value > threshold value set for back side
yes
no
car stop
accelerometer turns on
current value from accelerometer > threshold value set for accident detection
no
yes
accident detected
response module (GSM+GPS)
send current location to the saved phone number
if IR get signal
LED blink
end

8. PROGRAMMING

8.1 COMBINED PROGRAME

```
// defining the pins
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
#define c1 4
#define c2 5
#define c3 6
#define c4 7
//#define led 10
const int trigPin = 2;
const int echoPin = 3;
const int trigPin1 = 8;
const int echoPin1 = 9;
long duration;
int distance;
long duration1;
int distance1;
int flag=0;
int flag1=0;
void setup()
{lcd.begin();
lcd.backlight();
Serial.begin(9600);
pinMode( c1, OUTPUT);
pinMode( c2, OUTPUT);
pinMode( c3, OUTPUT);
pinMode( c4, OUTPUT);
digitalWrite( c1, LOW);
digitalWrite( c2, LOW);
digitalWrite( c3, LOW);
```

```
digitalWrite( c4, LOW);
  pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output
  pinMode(echoPin, INPUT); // Sets the echoPin as an Input
  pinMode(trigPin1, OUTPUT); // Sets the trigPin as an Output
  pinMode(echoPin1, INPUT); // Sets the echoPin as an Input}
void forward()
{digitalWrite( c1, LOW);
  digitalWrite( c2, HIGH);
  digitalWrite( c3,LOW);
  digitalWrite( c4, HIGH);
  Serial.println("forward");}
void backward()
{digitalWrite( c1, HIGH);
  digitalWrite( c2, LOW);
  digitalWrite( c3,HIGH);
  digitalWrite( c4, LOW);
  Serial.println("backward");}
void left()
{digitalWrite( c1, LOW);
  digitalWrite( c2, HIGH);
  digitalWrite( c3,LOW);
  digitalWrite( c4, LOW);
  Serial.println("left");}
void right()
{digitalWrite( c1, LOW);
  digitalWrite( c2,LOW);
  digitalWrite( c3,LOW);
  digitalWrite( c4, HIGH);
  Serial.println("right");}
void Stop()
{digitalWrite( c1, LOW);
  digitalWrite( c2, LOW);
  digitalWrite( c3,LOW);
  digitalWrite( c4, LOW);
  Serial.println("Stop");}
void loop()
```

```
{digitalWrite(trigPin1, LOW);
delayMicroseconds(2);
// Sets the trigPin on HIGH state for 10 micro seconds
digitalWrite(trigPin1, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin1, LOW);
// Reads the echoPin, returns the sound wave travel time in microseconds
duration1 = pulseIn(echoPin1, HIGH);
// Calculating the distance
distance1= duration1*0.034/2;
// Serial.print("distance from front ultrasonic");
//Serial.println(distance1);
//delay(500);
if (distance1>30)
{lcd.setCursor(0, 0);
lcd.print("path is clear:");
lcd.setCursor(14,0);
lcd.print(distance1);
//Serial.println("LED OFF");
//digitalWrite(led, LOW);
delay(500);
//lcd.clear();}
  else
  {lcd.setCursor(0, 0);
lcd.print("obstacle front:");
lcd.setCursor(14,0);
lcd.print(distance1);
//Serial.println("LED ON");
//digitalWrite(led, HIGH);
delay(500);}
   digitalWrite(trigPin, LOW);
   delayMicroseconds(2);
   // Sets the trigPin on HIGH state for 10 micro seconds
   digitalWrite(trigPin, HIGH);
   delayMicroseconds(10);
   digitalWrite(trigPin, LOW);//
```

```
Reads the echoPin, returns the sound wave travel time in microseconds
duration = pulseIn(echoPin, HIGH);// Calculating the distance
distance= duration*0.034/2;// Prints the distance on the Serial Monitor
  if (distance>20)
    {flag=0;
      lcd.setCursor(0, 1);
      lcd.print("path is clear:");
      lcd.setCursor(14,1);
      lcd.print(distance);}
    else if (distance<20 && flag1== 1)
    {flag=1;
      lcd.setCursor(0, 1);
      lcd.print("obstacle back:");
      lcd.setCursor(14,1);
      lcd.print(distance);
      Stop();
      flag1=0; }
    while (Serial.available()>0)
{char data=Serial.read();
if (data=='8')
{forward();}
if (data=='2' && flag==0)
{backward();
flag1=1;}
if (data=='4')
{left(); }
if (data=='6' )
{right();
}
if (data=='0' )
{Stop();
}}}
```

9. CONCLUSION

This report concludes with summary of automobile networking which we mainly focused on four main systems to minimize the road accidents in our day to day life and to increase a better awareness on road artifacts. By studying and understanding the reasons why most of the accidents occur is one of the reasons why we got motivated to find a solution for this problem and came up with this idea. This project is all about vehicle inbuilt technologies in which more and more systems can be included to overcome all the limitations.

The Inter- vehicular networking improves performance of the network in terms of transmission reliability, throughput, and interference by introducing cooperation among vehicles. A vehicle-to-vehicle communication code of conduct for collaborative collision warning, overtaking chances and location sensing. Rising wireless technologies for inter-vehicular communications are promising to dramatically decrease the number of fatal roadway accidents by providing early warnings.

10. REFERENCE

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